



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 26-200-1

Date: 15-3-2022

EDMS 26-200-0

**TECHNICAL SPECIFICATIONS
FOR
NEUTRAL GROUNDING RESISTOR
(NGR) (NGR) FOR COUPLING
TRANSFORMERS 22/11 KV WITH
RATING up to 30 MVA**

Issue: 3-2022 / Rev- 0

❖ توجد بنود اختيارية (Option items) يجب تحديدها بواسطة شركة التوزيع قبل الطرح.



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1. SCOPE

- This specification covers the minimum requirements for the design, manufacturing, testing and supply of Neutral Grounding Resistor (NGR) with a high current resistor.
- This NGR should be designed to absorb a large amount of energy without exceeding temperature limitation defined in the latest applicable standards.
- This NGR should be a high resistance grounding which limits ground fault current to the safe levels so that all the electrical equipment in the power system are protected and at the same time letting enough current to flow to operate the protective relays that will alarm or clear the faults.
- This NGR should be connected to the secondary side of the transformer (star side) between neutral point of the transformer and earth.

2. APPLICABLE STANDERDS

- In addition to the requirements of this specification, the latest revision of the following standards should be applicable for the equipment/material covered in this specification. In case of conflict, the manufacturer may propose equipment/material conforming to one group of Industry codes and standards hereunder without jeopardizing the requirements of this specification.

IEC 60071-1 Insulation Coordination Part-1: Definition, Principles and rules

IEC 60071-2 Insulation Coordination Part-2: Application Guide.

IEC 62271-200 A.C Metal Enclosed Switchgear and Control gear for Rated Voltage above 1 KV and up to and including 52 KV

IEC 60529 Degrees of Protection provided by Enclosures (IP code)

IEC 60137 Insulated Bushings for Alternating Voltages above 1000 Volts

IEC 60273 Characteristics of Indoor and Outdoor Post Insulators for Systems with Nominal Voltages greater than 1000 volts

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IEC 60168 Tests on Indoor and Outdoor Post Insulators of Ceramic Material or Glass for Systems with Nominal Voltages greater than 1000 volts.

IEC 60044-1 Instrument Transformers Part-1 Current Transformers

IEC 61869 Instrument transformers

IEEE 57.32 Standard Requirements, Terminology, and Test Procedure for Neutral Grounding Devices.

3. ENVIRONMENTAL CONDITIONS

- Minimum ambient temperature : -5°C.
- Maximum ambient temperature : 45°C (50 °c as option).
- Maximum relative humidity : ≤ 95%.
- Maximum altitude : 1000 m.

4. GENERAL SPECIFICATIONS

- System Rated voltage : 12-24 KV
- Service rated voltage : 11-22 KV
- Phase voltage : 6.36 - 12.72 KV
- Fault current : Refer to table 1
- Resistance value at 20 °C : Refer to table 2
- Tolerance : ± 10 %
- Fault time : 30 Sec
- Response time : ≤ 5 Sec
- Maximum Temperature rise : 760 °C
- One minute withstand voltage 50 Hz : 28 -50 KV
- BIL (Impulse withstand voltage) : 75 -128 KV

5. DESIGN CRITERIA

5.1.General

- The NGR should meet or exceed the requirements of this specification in all respects.
- The NGR should be designed for outdoor use.

5.2.Materials

5.2.1. Enclosure



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- The enclosure of the NGR should be made from hot dip galvanized steel sheets with thickness not less than 2 mm. and the galvanizing layer thickness should be not less than 55 microns.
- The degree of protection for the enclosure should be not less than IP23 without forced cooling.
- For coastal area, duplex system, which a combination of galvanizing and painting, should be used.
- The paint should be heat resistant to withstand high temperature not less than 760oC.
- The enclosure should have four lifting eyebolts on top with minimum hole diameter of 30 mm for balanced lifting.
- Earthing of the enclosure should be via two separate stud terminals fitted diagonally opposite each other with solder less connectors and suitable for termination of copper conductor
- For floor or steel plinth mounting, the enclosure should be provided with fixation holes.

5.2.2. Neutral Bushing

- The neutral bushing with copper stud in an air insulated cable box should be mounted on the side of the enclosure with undrilled removable gland plate and cover. The cable box should be suitable for terminating the incoming cable(s) and should house one current transformer or more. The CT(s) ratings as approved by ...EDC should be wired down to a terminal box mounted between 1.0 m to 1.5 m height from ground level.
- For neutral bushing mounted on the top cover (or as per ...EDC requirements) , the enclosure should be mounted on a plinth or structure to raise the height of the bushing 2.5 meters from the ground level to avoid accidental contact. The steel plinth should be part of supply.
- The NGR should have provision for connection of neutral of more than one transformer in parallel.
- The minimum creepage distance should not be less than 4 cm/KV.

5.2.3. Ground Terminal Bushing

- A ground terminal bushing with copper stud should be provided through the enclosure wall for the resistor earth connection to the substation earth mesh.

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- The connection between the resistor and earth mesh should be through a terminal with copper stud and should be suitable to accept hard drawn circular stranded copper conductors to form part of the earth grid mesh.
- The minimum creepage distance should not be less than 4 cm/KV

5.2.4. Earthing Resistor

- The NGR should be designed and tested to comply with IEEE standard C57.32-2015.
- The neutral earthing resistor should be of metal type suitable for outdoor service .
- The active part should be made of robust, non-corrosive, non-inductive, non-magnetic high temperature grid or coil type stainless steel metal alloy or stainless-steel or similar approved material capable of withstanding operating temperature conditions whilst retaining its strength.
- The resistor element material should be made from a nonmagnetic alloy with a low temperature coefficient of resistance to keep the value of the fault current stable for proper metering and relaying.
- The maximum temperature rise above ambient which will be attained by the active part of the resistor as a result of the flow of the rated continuous current under standard operating conditions, it may be expressed as an average hot spot temperature rise and it's limits as mentioned in IEEE C57.32-2015.
- The neutral cable from point of transformer to the earthing resistor should have a current carrying capacity of not less than the max current rating of the resistor.
- The earthing resistor could be separate or in corporate with the other neutral point equipment (current transformer & disconnectors, surge arrester.)
- Bolts, nuts and all internal and external ferrous parts should be hot dip galvanized steel or stainless steel.
- The resistor should be rated for 30 seconds unless .
- The ends of each resistor element made of stainless-steel alloy should be connected by welding or by nickel plated copper jumpers using two bolts per joint.

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- Each resistor elements on supporting bank should be locked by clamp and all electrical connections between elements should be made by double bolted copper connections.
- The resistor elements should be adequately supported on steel rods and porcelain insulators with minimum creepage distance not less than 4 cm/KV and designed to withstand the current flowing under fault conditions.
- The secondary support insulators should be designed and arranged to allow free expansion and contraction of the resistor assembly at operating temperature conditions without failure.

5.2.5. Current transformer

- The NGR should be equipped with a current transformer (CT) to protect the transformer against earth fault current and should has the following specifications:
 - CT Ratio : 300-600/5A
 - Class : 5p10
 - Burden : 10 VA
 - Characteristics : Refer to table 1& 2

Transformer rating (MVA)	5	10	15	20	25	30
Ground Fault current (A) for 11 KV system	200	400	600	800	1000	1200
Ground Fault current (A) for 22 KV system	100	200	300	400	500	600

Table 1

Transformer rating (MVA)	5	10	15	20	25	30
Resistance value (Ohm) at 20 °C for 11KV system	31.8	15.9	10.6	7.9	6.4	5.3
Resistance value (Ohm) at 20 °C for 22KV system	127.2	63.6	42.4	31.8	25.4	21.2

Table 2

- The current transformers should comply with IEC 60044-1 & IEC 61869-1&2.

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- The rated secondary current should be 5A.
- The detailed current transformer ratings e.g. secondary current, number of secondary windings, accuracy (5P10), burden, etc. required by ...EDC will be specified in the tender and the rated continuous current should be 1.2 In at least.
- The current transformers should be located directly next to the neutral point in order to ensure continuous feeding to the earth fault relays even in case of maintenance or repairing the neutral point equipment.
- Protection cores should be terminated on suitable terminals in the transformer cubicle for connection of wiring to protection relays.

5.2.6. Disconnecter

- The disconnecter should conform to the latest edition of IEC 62271-102 and should be designed to meet the exact single phase disconnecter as required by ...EDC, and it should be of outdoor type for vertical or horizontal mounting, and equipped with complete driving mechanism operated by D.C motor or manually.
- An external earthing stud connected to the metal base should be with enough size to accommodate twin lugged copper conductors with suitable cross-sectional area according to design calculation and IEC standard.
- The rated current of the disconnecter should be 630A.

5.2.7. Surge Arresters

- A surge arrester is required to control overvoltage when switching the associated transformer from the high voltage side.
- The surge arresters should be metal oxide type heavy duty, its housing should be made of porcelain or polymer insulators hermetically sealed.
- All metal parts of the surge arrester should be galvanized.
- Technical data for surge arresters should be as follows:

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Nominal voltage	11-22 KV
kJ/KV of rated voltage	2.5 kJ/KV
Nominal discharge current (8/20 μ s)	10 kA
Min. specific leakage path cm/KV	4 cm/KV

5.2.8. Control Wiring

- Control wiring should be of insulated standard copper conductor type of minimum cross- section 1.5 mm².
- Control wiring for use on CT secondaries should be of cross section 3mm².
- All control circuits and operating devices should be rated to withstand a voltage of 2 KV r.m.s. to earth for one minute.
- Wiring should be suitably protected in trunk or conduit within the cubicle.

6. TESTING

6.1. TYPE TEST

- All type tests prescribed in the relevant IEC and IEEE standards should be performed on the first unit of every new design, rating and size of NGR to be supplied to ...EDC.
- The following type tests, should be performed :
 1. Measurement of Degree of Protection of Enclosure.
 2. Temperature Rise Test.
 3. Measurement of Resistance Value at Stable Temperature.
 4. Lightning Impulse Test.

6.2. ROUTINE TEST

- All routine tests prescribed in the relevant IEC and IEEE standards, should be performed on each NGR.
- Test results should be provided for review and approval by ...EDC prior to delivery.
- The following routine tests should be performed:
 1. Visual and dimensional check to relevant drawings.
 2. Resistance measurement at ambient temperature.

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3. One-minute power frequency withstand voltage test 28 KV AC for 11kV system and 50 KV for 22 KV system .
4. Insulation resistance measurement before and after dielectric test 5 KV DC.
5. Galvanizing Test.

7. NAMEPLATE

- A nameplate should be fixed on the side of the NGR enclosure.
- The nameplate material should be stainless steel and should be fastened by stainless steel screws or rivets.
- The nameplate information should be stamped, or laser engraved into the nameplate.
- The information on the nameplate should be applied such that it is weatherproof, ultraviolet resistant, scratch resistant and permanent for the life of the NGR under normal handling and operating conditions. Silk-screened and laser etched paint are not acceptable.
- The nameplate information should include the following:
 1. Serial number.
 2. Type.
 3. Manufacturing year.
 4. Manufacturer's name or trademark.
 5. Name of manufacturing country.
 6. Applicable Standard
 7. No. of Resistor Frames
 8. Resistance (ohm) at 20 °C.
 9. Ground Fault current (A).
 10. Fault time (Seconds).
 11. System rated voltage (KV).
 12. Phase voltage (KV).
 13. Power frequency voltage (KV).
 14. BIL rating (KV).
 15. Rated frequency (Hz).
 16. Max temp. rise (°C).
 17. CT rating information (Ratio, Accuracy, Burden and Class)
 18. Total weight (Kg).



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19. Degree of protection (IP).

8. DRAWING AND CATALOGUE

- All necessary drawings including layout, dimensions, single line diagram of NGR should be submitted with the tender.
- All technical catalogues containing the technical data for all offered components of the unit should be submitted with the tender.

9. GUARANTEE TABLES

- Attached guarantee tables should be filled in and signed by the tenderer and enveloped with his tender.
- Any tender will not be accompanied with clear and complete guarantee tables should be rejected.



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**GURANTEE TABLE
FOR
NEUTRAL GROUNDING RESISTOR (NGR) FOR
COUPLING TRANSFORMERS 22/11KV WITH
RATING up to 30 MVA**

Maker 's name	
Standard specifications applied	
Country of manufacture	
Type or model	
System rated voltage	 KV
Phase voltage	 KV
Ground Fault current	A
Resistance value at 20 oC	Ohm
Resistance value Tolerance	 %
Enclosure protection class (IP)	
Enclosure thickness	mm
Galvanized layer thickness	micron
Fault time	Sec
Resistance type	
Temperature rise	C
Material of the resistive elements and parts active	
Thermal capacity	
Approx. net weight	Kg
CT Ratio	
CT Burden	VA
Dimensions of NGR :	
Length	Cm
Width	Cm
Height	 Cm

I/We guarantee the data given above for the equipment offered.

Signature:

Date: